

The LLM-Based AI Agent for College Students to Reduce Smartphone Addiction in the Morning

Qi Fang
Southern University of Science and
Technology
School of Automation and Intelligent
Manufacturing
Shenzhen, Guangdong, China
12332652@mail.sustech.edu.cn

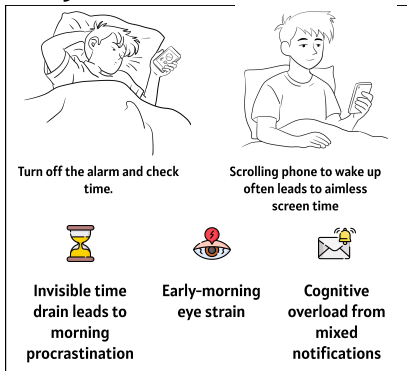
Qianwen Tan
Southern University of Science and
Technology
School of Automation and Intelligent
Manufacturing
Shenzhen, Guangdong, China
12332673@mail.sustech.edu.cn

Ziyan Xia
Southern University of Science and
Technology
School of Automation and Intelligent
Manufacturing
Shenzhen, Guangdong, China
12433036@mail.sustech.edu.cn

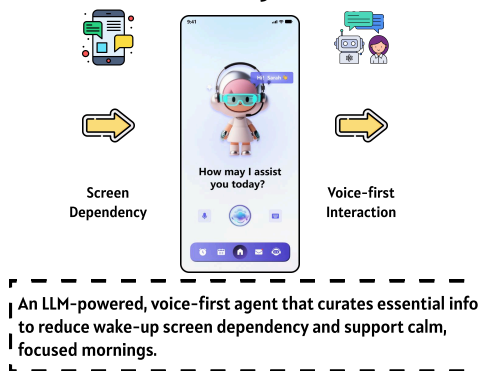
Kai Lin
Southern University of Science and
Technology
School of Automation and Intelligent
Manufacturing
Shenzhen, Guangdong, China
12431367@mail.sustech.edu.cn

Ziqian Bai*
Southern University of Science and
Technology
School of Automation and Intelligent
Manufacturing
Shenzhen, Guangdong, China
baizq@sustech.edu.cn

Background



LLM AI Agent



Storyboard

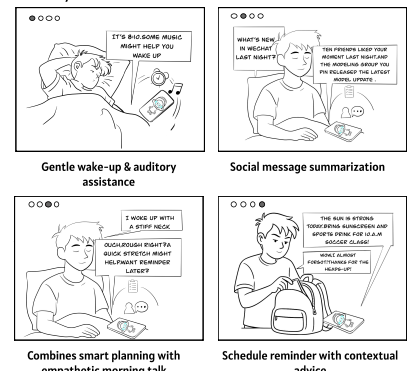


Figure 1: System Design of the AI Agent.

Abstract

Research shows that millions of people reach for their smartphones as soon as they wake up. The immediate influx of information and exposure to blue light can lead to mental fatigue, impaired cognitive function, and increased anxiety. Despite the extensive research conducted on smartphone addiction intervention techniques, there is a paucity of studies on LLM-Based intervention methods for smartphone addiction in the morning. This study aims to investigate the

*Ziqian Bai is the corresponding author.

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relationship between using smartphones first thing in the morning and emotional stress, with two primary research objectives: (1) developing a mathematical model characterizing the relationship between using smartphones first thing in the morning and emotional stress and (2) designing an LLM-Based AI Agent application that can provide features such as personalized integration and alerts of key smartphone information. By reducing the urgency to check smartphones upon waking, the agent aims to encourage healthier cognitive transitions into wakefulness and reduce stress-inducing digital behaviors.

CCS Concepts

• Human-centered computing → User interface programming.

Keywords

Smartphone Addiction, Morning Behavior, LLM-Based AI Agent, Artificial Intelligence

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1 Introduction

Research shows that millions of people check their smartphones for important information before they are out of bed [14]. Smartphone addiction in the morning has become a concern, especially for College students. Based on the 2019 statistics from the website BankMyCell, 87% of people check their phone within an hour of waking up or going to sleep. 69% of smartphone users check their phones within 5 minutes of waking up in the morning [10]. Some people use their smartphones as morning alarms and then start using them briefly (e.g. Checking messages or social media), which creates a habit of dependence [1]. For college students, this behavior is particularly pronounced, as they often receive a high volume of academic and social information.

This study has two primary objectives: (1) constructing a mathematical model to quantitatively characterize the relationship between morning smartphone overuse and emotional stress levels, and (2) designing an AI Agent based on Large Language Models (LLMs). This AI Agent will provide personalized information integration and intelligent alerts for critical smartphone notifications, thereby reducing users' urge to compulsively check their phones as soon as they wake up. This system aims to mitigate stress-inducing digital behaviors and promote improved psychological well-being by facilitating healthier cognitive transitions from sleep to wakefulness.

2 Related Work

2.1 Existing Applications Interventions to Reduce Smartphone Addiction

Interventions for Smartphone addiction can be broadly categorized into six distinct categories [4]: psychotherapy, cognitive training, behavioral intervention, application restriction, social intervention, and complementary and alternative medicine. App restriction is a smartphone-based intervention that may be a cost-effective way to reduce students' smartphone addiction [7]. A lot of research has been done on designing digital tools and techniques to reduce people's smartphone addiction: (1) Limiting the use of applications [5, 9]; (2) Enhancing self-awareness by monitoring and informing users of their usage statistics [6, 8]; (3) Distraction or group supervision [2, 13]. Existing interventions have shown some success in enhancing user self-awareness and reducing usage time, but often rely on simplistic criteria and focus broadly on general addictive behaviors, overlooking specific usage patterns such as morning use. The LLM-Based intervention system has significant advantages when it comes to analysing specific situations. By using deep learning and natural language processing, it can provide more personalised, context-aware strategies by analysing individual behavior patterns and preferences.

2.2 Existing LLM-Based Interventions to Reduce Smartphone Addiction

There has been some initial research using LLMs to deliver interventions. LLMs are able to analyze massive amounts of data and identify patterns that may not be captured by humans. In addition, users can provide feedback to the AI, enabling LLMs to improve their predictions and personalize interventions based on individual needs and behaviors. MindShift [12] reduces usage and increases perceived control by aligning interventions with mental states. However, users criticise its rigid, templated approach. StayFocused [3] uses goal-reflection prompts and a chatbot to reduce screen time and increase self-awareness, but its limited functionality and poor contextual understanding negatively impact the user experience. There is not much research on using LLMs to address smartphone addiction, and this is an area that requires further exploration.

3 Design Opportunities

Reducing smartphone addiction among college students in the morning is a critical issue. Existing studies have shown benefits in addressing smartphone addiction but still have some limitations: (1) Research into smartphone addiction mostly ignores the morning stage, which plays a key role in the behavior of college students; (2) Existing software interventions do not design for smartphone addiction among college students in the morning, and have not fully explored the relationship between using smartphones first thing in the morning and emotional stress among college students; (3) Existing software interventions do not make good use of the contextual capabilities of LLM to personalise intervention design.

Despite these limitations, this study emphasizes its significance by focusing on the specific issue of morning smartphone addiction among college students. This study aims to develop better understanding of the behavioral patterns and emotional stress associated with using smartphones first thing in the morning. This study will conduct semi-structured interviews and questionnaires among college students to explore the specific causes of morning smartphone addiction. In addition, this study will conduct a comprehensive analysis by integrating the StudentLife dataset [11] with the new datasets that this study plans to collect in the future. By analyzing these data, this study will construct a mathematical model based on logistic regression to quantify the relationship between the frequency of checking smartphones upon waking up in the morning and emotional stress.

4 Exploring Users' Behavioral Patterns

4.1 Semi-structured Interview Studies

This study conducted interviews with five college students to explore their smartphone usage patterns and potential needs upon waking. Results revealed a common behavioral sequence: turning off the alarm, immediately checking social messages, and then browsing entertainment apps. Social platforms such as WeChat (a Chinese App like Facebook) and QQ (a Chinese App like WhatsApp) play a central role in morning information processing. Although users tend to assign subjective priority levels to messages, they often struggle with information overload and disruptions caused by unexpected tasks.

All respondents use their smartphones to make themselves more awake. People mainly use short-video apps like TikTok, and text-based apps like Weibo and news websites. However, this often led to feelings of wasted time and habitual screen dependency. The interviews identified core requirements that users have for a voice-based AI assistant: (1) Users expect the assistant to extract messages automatically, update their schedule and detect any disagreements. These information should be delivered through a combination of voice and text to reduce the burden on the brain and memory problems during the morning; (2) Maintaining user control over prioritization and scheduling is essential; (3) To address issues such as time delays and anxiety caused by screen time, users supported including personalized multimodal content, such as curated music, podcasts, or conversational voice support, as an alternative to unconscious smartphone use; (4) Regarding interaction design, users generally prefer that the assistant launch automatically after the alarm is turned off; (5) They also stressed the need for context-aware privacy protection, such as switching to vibration and text summaries in public settings.

4.2 Questionnaire Surveys

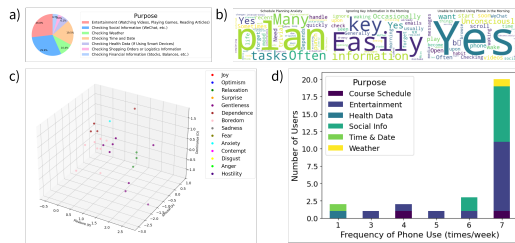


Figure 2: a) Primary reasons for checking the phone after waking up; b) Behavioral tendencies of using smartphones first thing in the morning; c) Distribution of Emotions in PAD 3D Space; d) Relationship between mood and frequency of using smartphones first thing in the morning.

To explore (1) whether using smartphones first thing in the morning affects users' emotional stress, and (2) the purpose of using smartphones first thing in the morning. This study designed a questionnaire to collect data from 29 college students (aged 19–27). This study employed the PAD effective measure. The Euclidean distance algorithm was used to calculate the distance between users' emotional tendencies and 14 basic emotions. The questionnaire (Figure 2) shows that people mainly use phones upon waking to check social media, entertain themselves, or view the time, date, and weather. These insights will inform the software prototype design. Moreover, daily phone use upon waking (7 times/week) is linked to higher levels of negative emotions like anxiety, boredom, sadness, and dependency.

5 System Design

Based on the objective of this study and the findings of Sections 4.1 and 4.2, we designed an LLM-Voice-Based AI Agent. The core contributions of the AI Agent are (1) integrating overnight messages from social media platforms, then filtering and broadcasting the

most relevant information via voice; (2) managing schedules via voice commands like "Remind me of a meeting at 2 PM today" or "Record a to-do item," reducing manual input hassle and promoting a hands-free, efficient morning routine; (3) using the contextual capabilities of LLM to understand and integrate with calendar and task management systems to update schedules and reminders; (4) waking users gently with voice conversations, then offering time, weather, and so on. It also recommends activities like meditation or yoga based on user preferences, helping avoid distractions and improving morning efficiency. Development will follow a phased strategy, starting with rapid prototyping and refining based on user feedback.

5.1 Functional and System Design

This software prototype is designed with the following main functional modules: (1) Personality Settings, which allows users to customize AI Agent and message notification methods; (2) Unread message checking module, which will summarize the user's new messages and output key messages received by the user's smartphone in the form of conversational speech; (3) Schedule Checking Module, which can rationalize time planning based on the user's information; (4) Morning Guidance Module, which can provide suggestions for morning activities and dress code based on weather and more; (5) Procrastination Management Module, which can remind the user to complete tasks and intervene when they are not completed. This AI Agent uses a modular, voice-first architecture. Its Context-Aware Prompting engine tailors interactions to the user's state and environmental cues in real time.

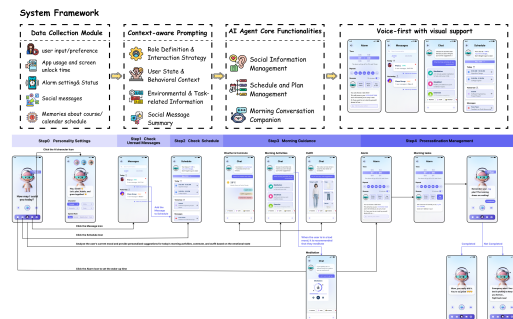


Figure 3: Framework of the AI Agent.

5.2 Intervention Trigger Mechanism

To achieve a voice interaction experience that is more attuned to the college students' morning state, this study introduces a Context-aware Prompting (CAP) mechanism and a voice interaction-based intervention triggering mechanism.

5.2.1 Context-aware Prompting (CAP) Mechanism. The CAP mechanism will continuously monitor post-wake user behavior. This includes screen unlock time, first app usage, cumulative usage duration, and alarm settings (time, ringing, and dismissal) to assess waking states and behavioral patterns. This mechanism uses course timetables and information about how well students are doing in

their studies to create short, clear suggestions that consider a student's behavior, emotions, and the environment. These prompts guide the AI Agent's tone and how it help in real time, going beyond the simple responses of regular assistants.

5.2.2 Voice Interaction-based Intervention Triggering Mechanism. This mechanism aims to reduce the user's dependence on the smartphone screen: (1) Integrating the information received during the user's sleep period, and when the user wakes up, the AI Agent will output the integrated information via voice; (2) Waking up the user with a soft voice dialog (or music, blog, radio, etc.); (3) Users can set schedules through voice commands; (4) Offering morning suggestions (such as activity recommendations, weather reminders and so on) via voice; (5) When the user fails to complete the task on time, the AI Agent will automatically trigger a voice alert to remind the user to continue.

6 Usability Test

We conducted usability tests and evaluations on the designed prototype and received suggestions on how to improve it. The demographic data of the participants includes 3 male and 3 female college students whose age is between 18 and 27 years. We adopt a test method: A series of prompt words were set in advance according to the test tasks, allowing users to make voice calls through Cici (AI app), simulating the morning usage scenario of the AI Agent, and asking users to rate the functional modules (0-5 points).

The results of this usability test show that prototype performs well. The users think that voice interaction is convenient and efficient. AI can accurately identify and respond to users' emotions, showing that it is intelligent and human-like. The combination of text and voice reminders is effective. During this test, the subjects made some suggestions for improvement: (1) To optimize the information filtering mechanism to improve the accuracy of the summary; (2) To increase the review of completed tasks and encourage feedback to enhance the interactive experience; (3) It is expected that the AI can identify the emotional changes in more subtly based on the daily behavioral patterns and provide more intimate emotional support; (4) To add varies of voice styles for the user to select, which can enhance the user's experiences.

7 Conclusion

This study explores the potential of AI Agent based on LLMs to address morning smartphone addiction and emotional stress. In the preliminary research stage, we have identified some insights: (1) Most students habitually use their smartphones immediately upon waking, but they do not prioritise received messages rationally; (2) College students engage in recreational phone use to increase alertness, but this can cause screen fatigue, pressure and negative emotions such as anxiety. Finally, usability testing with 6 students provided valuable feedback for improving the prototype. Future work will focus on enhancing the context-awareness and personalization capabilities of AI agents by integrating and analyzing more datasets. Additionally, we will setup new datasets to conduct longitudinal studies that will evaluate the long-term impact of AI agents in reducing morning smartphone addiction and optimizing the users' daily routine.

Acknowledgments

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References

- [1] Istiqoma and Ahmad Yasser Effendi. 2019/11. Mobile Phone Addiction: Smartphone Usage among Digital Natives With Disabilities In Indonesia. In *Proceedings of the 2nd International Conference on Strategic and Global Studies (ICSGS 2018)*. Atlantis Press, Jakarta, Indonesia, 90–98. doi:10.2991/icsgs-18.2019.13
- [2] Jaejeung Kim, Joonyoung Park, Hyunsoo Lee, Minsam Ko, and Uichin Lee. 2019. LocknType: Lockout Task Intervention for Discouraging Smartphone App Use. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland Uk) (CHI '19). Association for Computing Machinery, New York, NY, USA, 1–12. doi:10.1145/3290605.3300927
- [3] Zhuoyang Li, Minhui Liang, Ray Lc, and Yuhuan Luo. 2024. StayFocused: Examining the Effects of Reflective Prompts and Chatbot Support on Compulsive Smartphone Use. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 247, 19 pages. doi:10.1145/3613904.3642479
- [4] Xiao Xia Liu. 2021. A systematic review of prevention and intervention strategies for smartphone addiction in students: Applicability during the COVID-19 pandemic. *Journal of Evidence-Based Psychotherapies* 21, 2 (2021), 3–36. <https://doi.org/10.24193/jebp.2021.2.9>
- [5] Kai Lukoff, Ulrik Lyngs, Himanshu Zade, J. Vera Liao, James Choi, Kaiyue Fan, Sean A. Munson, and Alexis Hiniker. 2021. How the Design of YouTube Influences User Sense of Agency. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (CHI '21). Association for Computing Machinery, New York, NY, USA, Article 368, 17 pages. doi:10.1145/3411764.3445467
- [6] Ulrik Lyngs, Kai Lukoff, Laura Csuka, Petr Slovák, Max Van Kleek, and Nigel Shadbolt. 2022. The Goldilocks level of support: Using user reviews, ratings, and installation numbers to investigate digital self-control tools. *International Journal of Human-Computer Studies* 166 (2022), 102869. doi:10.1016/j.ijhcs.2022.102869
- [7] Saoirse Mac Carthaigh. 2020. The effectiveness of interventions to reduce excessive mobile device usage among adolescents: A systematic review. *Neurology, Psychiatry and Brain Research* 35 (2020), 29–37. doi:10.1016/j.nprb.2019.11.002
- [8] Alberto Monge Roffarello and Luigi De Russis. 2019. The Race Towards Digital Wellbeing: Issues and Opportunities. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland Uk) (CHI '19). Association for Computing Machinery, New York, NY, USA, 1–14. doi:10.1145/3290605.3300616
- [9] Adiba Orzikulova, Hyunsung Cho, Hye-Young Chung, Hwajung Hong, Uichin Lee, and Sung-Ju Lee. 2023. FinerMe: Examining App-level and Feature-level Interventions to Regulate Mobile Social Media Use. *Proc. ACM Hum.-Comput. Interact.* 7, CSCW2, Article 274 (Oct. 2023), 30 pages. doi:10.1145/3610065
- [10] Ash Turner. 2020. How Many Phones Are in the World? BankMyCell. Retrieved April 29, 2021, from <https://www.bankmycell.com/blog/how-many-phones-are-in-the-world>.
- [11] Rui Wang, Fanglin Chen, Zhenyu Chen, Tianxing Li, Gabriella Harari, Stefanie Tignor, Xia Zhou, Dror Ben-Zeev, and Andrew T. Campbell. 2014. StudentLife: assessing mental health, academic performance and behavioral trends of college students using smartphones. In *Proceedings of the 2014 ACM International Joint Conference on Pervasive and Ubiquitous Computing* (Seattle, Washington) (UbiComp '14). Association for Computing Machinery, New York, NY, USA, 3–14. doi:10.1145/2632048.2632054
- [12] Ruolan Wu, Chun Yu, Xiaole Pan, Yujia Liu, Ningning Zhang, Yue Fu, Yuhuan Wang, Zhi Zheng, Li Chen, Qiaolei Jiang, Xuhai Xu, and Yuanchun Shi. 2024. MindShift: Leveraging Large Language Models for Mental-States-Based Problematic Smartphone Use Intervention. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 248, 24 pages. doi:10.1145/3613904.3642790
- [13] Xuhai Xu, Tianyuan Zou, Han Xiao, Yanzhang Li, Ruolin Wang, Tianyi Yuan, Yuntao Wang, Yuanchun Shi, Jennifer Mankoff, and Anind K Dey. 2022. TypeOut: Leveraging Just-in-Time Self-Affirmation for Smartphone Overuse Reduction. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (CHI '22). Association for Computing Machinery, New York, NY, USA, Article 442, 17 pages. doi:10.1145/3491102.3517476
- [14] Heng Yue, Xuemin Zhang, Junfang Sun, Min Liu, and Hujieletu Bao. 2021. The relationships between negative emotions and latent classes of smartphone addiction. *PLoS ONE* 16, 3 (2021), e0248555.